

RAMCO INSTITUTE OF TECHNOLOGY
Department of Electronics and Communication Engineering
Academic Year: 2023- 2024 (Odd Semester)

Active Learning Practices

Degree, Semester & Branch: V Semester B.E.ECE A & B

Course Code & Title: CEC352 Satellite Communication

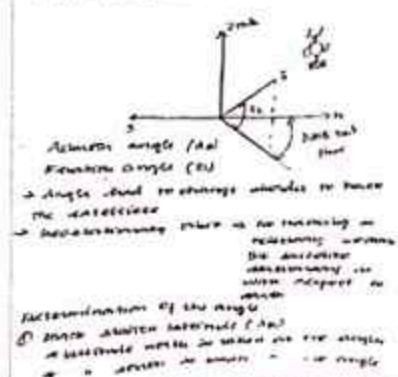
Name of the Faculty member: Ms.R.Chandralekha

Active Learning Practices Execution

UNIT I SATELLITE ORBITS

Activity: One Minute Paper
Topic: Look Angle Determination

Look angle determination:-
A direction should be in the satellite



- ② Given station longitude (λ)
is longitude when the station is at the angle
is longitude when the station is at the angle
- ③ Longitude of station longitude point (λ)
or satellite longitude
- For look angle calculation,
station radius is 6370 km
R = 6370 km
R is not at station point negative



Signature
1/8/23
Signature of the faculty

Signature
1/11/23
HOD

RAMCO INSTITUTE OF TECHNOLOGY
Department of Electronics and Communication Engineering
Academic Year: 2023- 2024 (Odd Semester)

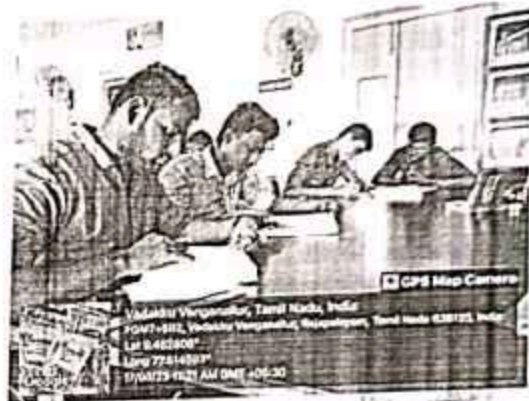
Active Learning Practices

Degree, Semester & Branch: V Semester B.E.ECE A & B
Course Code & Title: CEC352 Satellite Communication
Name of the Faculty member: Ms.R.Chandralekha

Active Learning Practices Execution

UNIT II SPACE SEGMENT

Activity: Open Book Test (Library Visit)
Topic: Telemetry, Tracking and Command Subsystem




22/8/23
Signature of the faculty


11/11/23
HOD

RAMCO INSTITUTE OF TECHNOLOGY
Department of Electronics and Communication Engineering
Academic Year: 2023- 2024 (Odd Semester)

Active Learning Practices

Degree, Semester & Branch: V Semester B.E.ECE A & B

Course Code & Title: CEC352 Satellite Communication

Name of the Faculty member: Ms.R. Chandralekha

Active Learning Practices Execution

UNIT III SATELLITE LINK DESIGN

Activity: Zero-minute speech

Topic: Basic link analysis – Downlink analysis




14/9/23

Signature of the faculty



HOD

RAMCO INSTITUTE OF TECHNOLOGY
Department of Electronics and Communication Engineering
Academic Year: 2023- 2024 (Odd Semester)

Active Learning Practices

Degree, Semester & Branch: V Semester B.E.ECE A & B

Course Code & Title: CEC352 Satellite Communication

Name of the Faculty member: Ms.R.Chandralekha

Active Learning Practices Execution

UNIT IV SATELLITE ACCESS AND CODING

Activity: Jigsaw

Topic: Code Division Multiple Access (CDMA)




Signature of the Faculty


HOD

RAMCO INSTITUTE OF TECHNOLOGY
Department of Electronics and Communication Engineering
Academic Year: 2023- 2024(Odd Semester)

Active Learning Practices

Degree, Semester & Branch: V Semester B.E.ECE A & B

Course Code & Title: CEC352 Satellite Communication

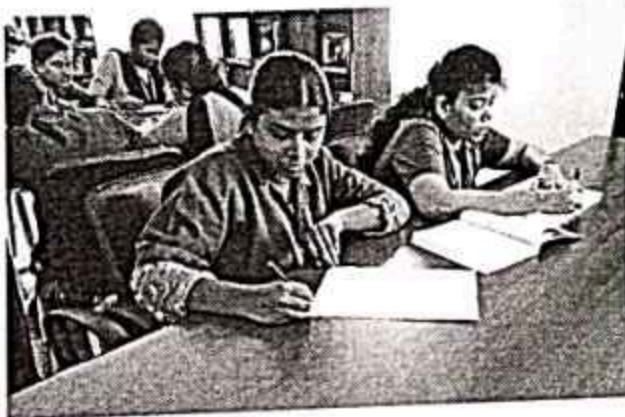
Name of the Faculty member: Ms.R. Chandrakha

Active Learning Practices Execution

UNIT V SATELLITE APPLICATIONS

Activity: Reflective Journal (Library Visit)

Topic: GPS – Position Location Principles




Signature of the faculty


HOD



RAMCO INSTITUTE OF TECHNOLOGY

Approved by AICTE, New Delhi & Affiliated to Anna University
Accredited by NAAC & An ISO 9001:2015 Certified Institution
NBA Accredited UG Programs: CSE, EEE, ECE and MECH

Department of Electronics and Communication Engineering

Academic Year 2023-2024 (Odd Semester)

Planning Document

Degree, Semester & Branch: V Semester B.E. ECE A & B

Course Code & Title: CEC352 Satellite Communication

Name of the Faculty member: Ms.R.Chandralekha, AP/ECE

- Unit/Topic: Unit-IV / Code Division Multiple Access (CDMA)
- Course Outcome: CO4
- Topic Learning Outcome: TLO 14
- Activity Chosen: Jigsaw

Learning Outcomes:

The Student will be able to

- ❖ Understand the techniques used in CDMA
- ❖ Know the interference effects in CDMA
- ❖ Participate actively in the learning process
- ❖ Discuss the concepts, convey idea and share the views of synchronization, codeword, power efficiency and the requirement of guard band and guard times.
- ❖ Communicate effectively by sharing their views among the teammates

Justification:

The basic concept of multiple access scheme was already learnt in Wireless Communication. The topic is related to the same concepts and easy to understand. The Jigsaw Strategy is an efficient way to learn the course material in a cooperative learning style. The jigsaw process encourages listening, engagement, and empathy by giving each member of the group an essential part to play in the academic activity. The activity 'JIGSAW' is used to recall the concepts learnt and enhance the learning through this collaboration

- Time Allotted for the Activity: 50 Minutes

- Details of the Implementation:

(i) Materials for the Activity:

The materials for the preparation of the students will be shared one week before through LMS Canvas. Including the material, students were asked to refer to the web content also.

References:

Book Title/ Author/ Publisher/ Edition	Page No.
1. Dennis Roddy, "Satellite Communication", 4th Edition, Tata McGraw Hill International, 2006.	472-483
Websites	
1. https://ocw.mit.edu/courses/6-450-principles-of-digital-communications-i-fall-2006/resources/lecture-24-case-study-on-code-division-multiple-access/	
2. https://www.jigsaw.org/	
3. https://www.newcastle.edu.au/data/assets/pdf_file/0016/109600/Jigsaw-	

(ii) Formation of Groups:

Student groups were created as below:

- Class Strength: 60
- Number of groups: 06
- Members per group: $06(7) + 05(2) + 04(2)$

• Plan for Implementing this Activity:

- 8 groups are formed with 4 or 5 members
- Each group is allotted with a name and its known as Home group.
- Each expert group is allotted with one topic
 - Direct sequence spread spectrum
 - The code signal $c(t)$
 - Acquisition and tracking
 - Spectrum spreading and dispreading
 - CDMA throughput
- The expert group members discuss the topic in detail for 20 minutes with the learning materials already posted in the course website – canvas.
- The Expert group members should go to their original shape group and discuss the points (30 Minutes) to the other members.
- All the members in home group learnt about all the topics through expert group members.
- Finally one from each group should summarize the points of the topic Fixed and Floating point representation to the class (20 Minutes).
- Review of points and conclusion of the activity (5 minutes)
- Thus the concept of CDMA is known to all by involving the students actively

Expected Difficulties:

- Lack of preparation: Making the students to learn the materials is the challenging task.
- Each student has to read the material posted in the course website.
- Non participation in the activity: All the students should be made involve in the activity
- Time management: The Jigsaw activity should be able to complete in the planned duration.

Plan for preventing these difficulties:

- Making each student accountable by making formative and summative assessment for individual and group performance.
- Including the assessment mark for final internal mark calculation makes each student to participate actively in the Jigsaw activity.
- Each student contribution mark in the assessment will be based on the discussions in the course website and whatsapp group through which all the students will participate in the activity.
- Posting announcements periodically for learning and preparing the presentation.
- The Time management will be avoided by using timer clock for 10 minutes duration.

• Student Feedback: Written Feedback

1. How do you rate the activity for learning the concept? Give tick mark (✓)			
Excellent	Good	Satisfactory	Poor
	✓		
2. How do you rate the reference material provided related to the topic? Give tick mark (✓)			
Excellent	Good	Satisfactory	Poor
	✓		
3. Have you enjoyed the activity? Give tick mark (✓)			Yes: ✓ No:
4. Challenges faced during activity/presentation. Give tick mark (✓)			Yes: No: ✓
If yes what?:			
G1-SURESH KUMAR G1-Suresh 17.10.23 Name & Signature of the Student with date			


 Signature of Faculty Member


 HOD